Leakage-Aware Deep Learning for Financial Event-State Forecasting

Jose Ignacio Valdivia Aguero

University of Luxembourg

DL Course Project

Multi-horizon excess-return forecasting with leakage-aware deep-learning benchmarks.

Talk Roadmap

Question

- Can aligned company-news event states improve multi-horizon excess-return forecasts?
- Can deep models help without leaking future information?

Answer

- XGBoost is the strongest lockbox RMSE/MAE forecaster.
- TSMixer is the strongest lockbox portfolio-ranking model.
- The main contribution is a rigorous, reproducible forecasting system.

Why This Is Hard

- Returns have low signal-to-noise and strong nonstationarity.
- News can matter, but timestamps, no-news days, and provider truncation create hidden biases.
- Model selection can overfit validation folds even when deep models look plausible.

Design principle

Train/validation is for search. Lockbox is the final excess-return forecasting test.

Metric principle

ROC-AUC is not applicable: this is continuous excess-return regression.

Protocol: Leakage-Aware Artifact Pipeline

Stage	Artifact	Key control	Output
Data	Top-50 + OHLCV + news	no raw overwrite	row universe
Features	market + event states	no-news rows preserved	sidecars
Splits	expanding folds + lockbox	purge/embargo	split manifest
Training	fold-local preprocessing	no lockbox tuning	saved preds
Evaluation	RMSE/MAE + rank IC	bootstrap + lockbox	reports

- Horizons: 1d, 5d, 10d, 20d, 60d.
- Lockbox ends 2026-02-13; all result roots keep manifests and prediction packets.

State of the Art Context

Financial ML baselines

- Efficient-market discipline and expected-return multiple testing.
- Gradient boosting is still a very strong tabular baseline.
- Forecast metrics: RMSE, MAE, OOS R^2, rank IC.

Deep learning candidates

- FinBERT/BERT for semantic financial text features.
- TSMixer and Transformer-family sequence models.
- PatchTST, iTransformer, TFT, and Chronos motivate richer temporal architectures.

Implemented Model Stack

Text/Event

- Alpha Vantage aligned news.
- No-news as explicit state.
- FinBERT sidecars for later fusion.

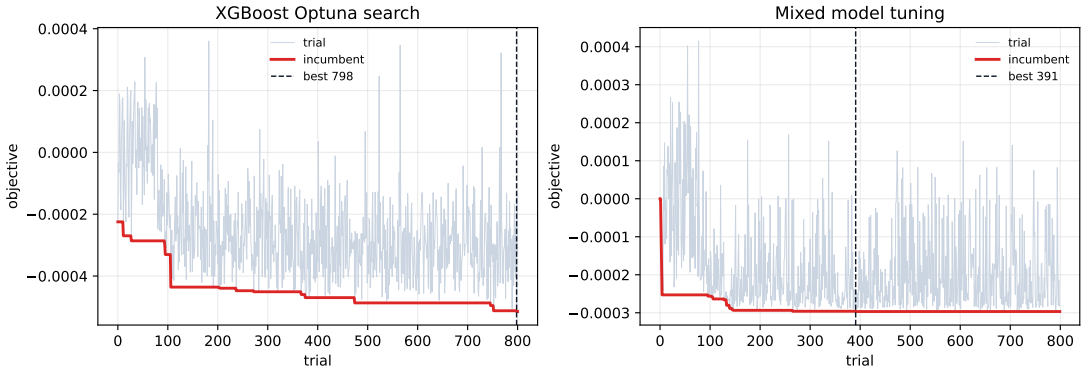
Market

- Lag/rolling features.
- Linear and fixed XGBoost baselines.
- XGBoost: 800-trial tuned baseline.

Sequence/Mixer

- TSMixer sequence models.
- Readout correction audit.
- Mixed model: frozen-base horizon mixer.

Tuning Trace: Search Was Auditable



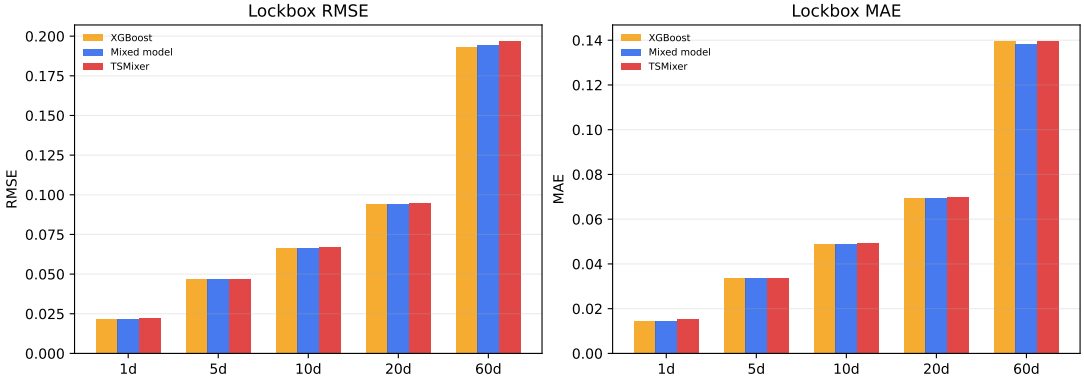
- XGBoost Optuna search completed 800 trials; selected model wins lockbox RMSE/MAE.
- Mixed model completed 800 meta/mixer trials; validation gains did not beat XGBoost error on lockbox.
- Both searches used zero lockbox rows for tuning.

Exact Lockbox RMSE by Horizon

Horizon	XGBoost	TSMixer	Mixed model	Best
1d	0.0214	0.0221	0.0214	XGBoost
5d	0.0468	0.0469	0.0469	XGBoost
10d	0.0661	0.0670	0.0665	XGBoost
20d	0.0943	0.0949	0.0944	XGBoost
60d	0.1929	0.1972	0.1942	XGBoost

- Error increases sharply with horizon, as expected for longer excess-return windows.
- XGBoost is the most consistent lockbox error-metric model.

Primary Metrics: Lockbox RMSE and MAE



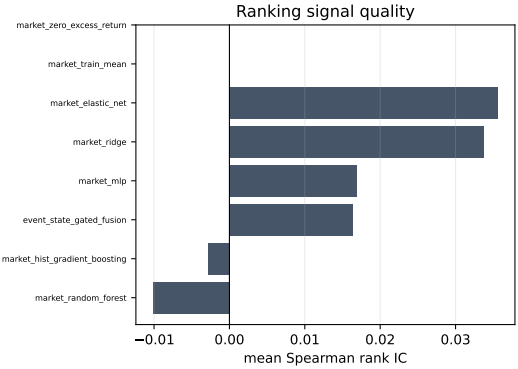
- Regression metrics are primary because labels are continuous forward excess returns.
- XGBoost gives the strongest lockbox RMSE profile across horizons.

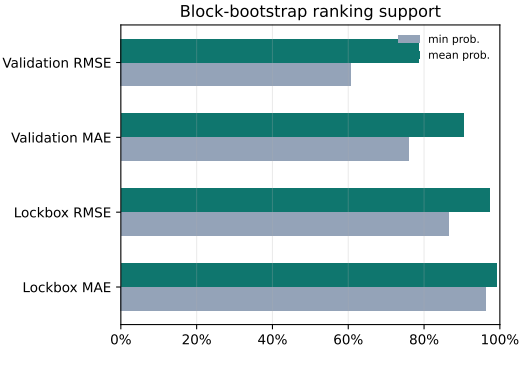
Error Winner vs Ranking Winner

Question	Winner	Evidence
Lowest lockbox RMSE/MAE	XGBoost	best RMSE across all 5 horizons
Best lockbox portfolio simulation	TSMixer	15.05%, \$115,045 over 137 days
Best Mixed model role	validation fusion	validation delta -2.97×10^{-4} vs XGB
Mixed lockbox error vs XGB	XGBoost	Mixed delta $+9.20 \times 10^{-5}$
Main interpretation	objective matters	calibrated error and ranking diverge

- XGBoost learned the most accurate excess-return levels under RMSE/MAE.
- TSMixer produced the strongest cross-sectional ranking signal in portfolio simulation.

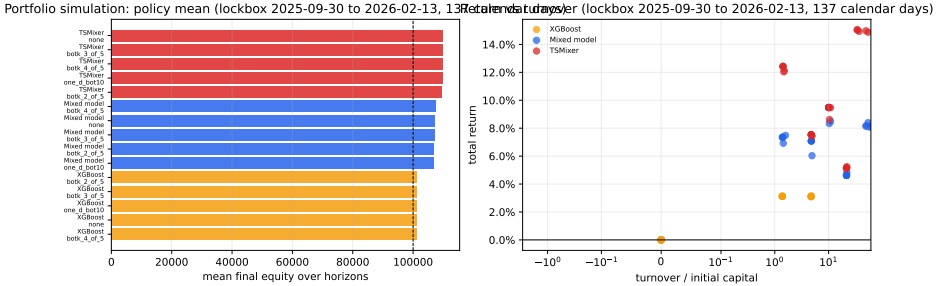
Secondary Evidence: Ranking and Uncertainty





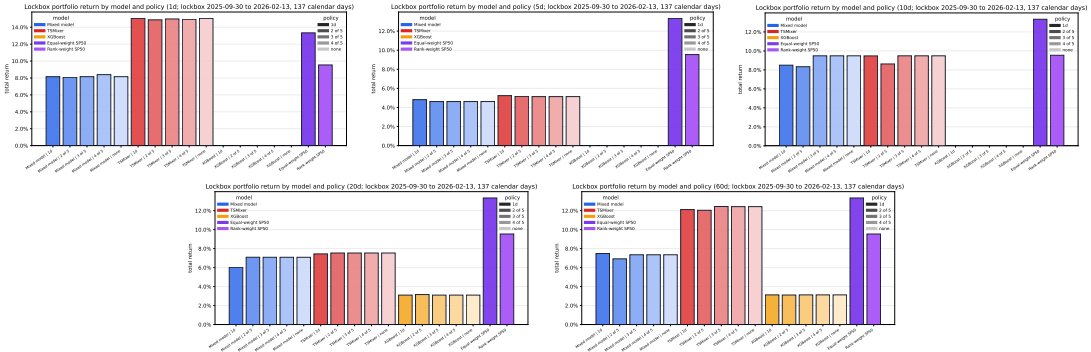
- Rank IC and hit-rate results describe cross-sectional forecast quality.
- Moving-block bootstrap checks temporal dependence and uncertainty.

Lockbox Portfolio Simulation



- Initial capital: $100k; long-only, fractional shares, no costs, no slippage, no taxes.
- Lockbox trading window: 2025-09-30 to 2026-02-13, 137 calendar days, about 4.5 months.
- Best row: $115,045 final equity; it uses frozen TSMixer sequence predictions.
- XGBoost short-horizon no-trade cases identify flat-rank signal construction for the next model.

Portfolio Bar Charts by Horizon



- Bars include model-policy simulations plus Equal-weight SP50 and Rank-weight SP50 proxies.
- The 15.05% top row is over 137 calendar days, not a long annual test.

Main Numerical Results

Evidence	Result	Interpretation
XGBoost lockbox delta	-5.34×10^{-4} vs prior XGB	best error forecaster
XGBoost both-metric rate	62.2%	consistent error improvement
TSMixer best portfolio row	15.05%, \$115,045	best ranking simulation
Mixed validation delta	-2.97×10^{-4} vs XGB	looked useful in search
Mixed lockbox delta	$+9.20 \times 10^{-5}$ vs XGB	fusion did not beat XGBoost error
ROC-AUC	not reported	continuous regression target

How Good Is the Final Model?

Dimension	Judgment	Evidence
Error forecasting	XGBoost leads	best RMSE across all horizons
Relative error gain	measurable	XGBoost delta -5.34×10^{-4}
Ranking simulation	TSMixer leads	15.05% top row over 137 days
Mixed model value	useful validation fusion	-2.97×10^{-4} validation delta
Model-selection lesson	objective-specific winner	error and ranking diverge
Research value	strong	leakage-aware artifacts and audits

Discussion: What I Learned

Positive lessons

- Reproducibility can be engineered through manifests, saved predictions, and tests.
- Tree baselines are hard to beat in small financial tabular settings.
- Deep sequence models need architecture audits and strong baselines; high validation score does not automatically translate to lockbox performance.

Next steps

- Try a deeper reinforcement learning model with capacity to rank predictions from most probable to least probable.
- Expand data collection beyond API limits with a reproducible scraping/cache layer.
- Add transaction costs and slippage before any portfolio-style interpretation.

Final Conclusion

Scientific conclusion

XGBoost is the final strongest lockbox error forecaster; TSMixer is the strongest lockbox ranking-simulation model.

Deep learning conclusion

For financial forecasting, the most important DL competence is not just adding BERT, TSMixer, or Transformers; it is controlling leakage, validating architecture semantics, and resisting overfit narratives.

Performance and simulation conclusion

Performance gains are measurable across the lockbox: XGBoost improves calibrated excess-return error, while TSMixer gives the best realized portfolio-ranking simulation at 15.05% over 137 calendar days.

THANK YOU